

Censored GP Reconstruction Experiments

Summary since last meeting: 1D uncertainty, lengthscale tuning, 2D shapes, observation density, and kernel priors

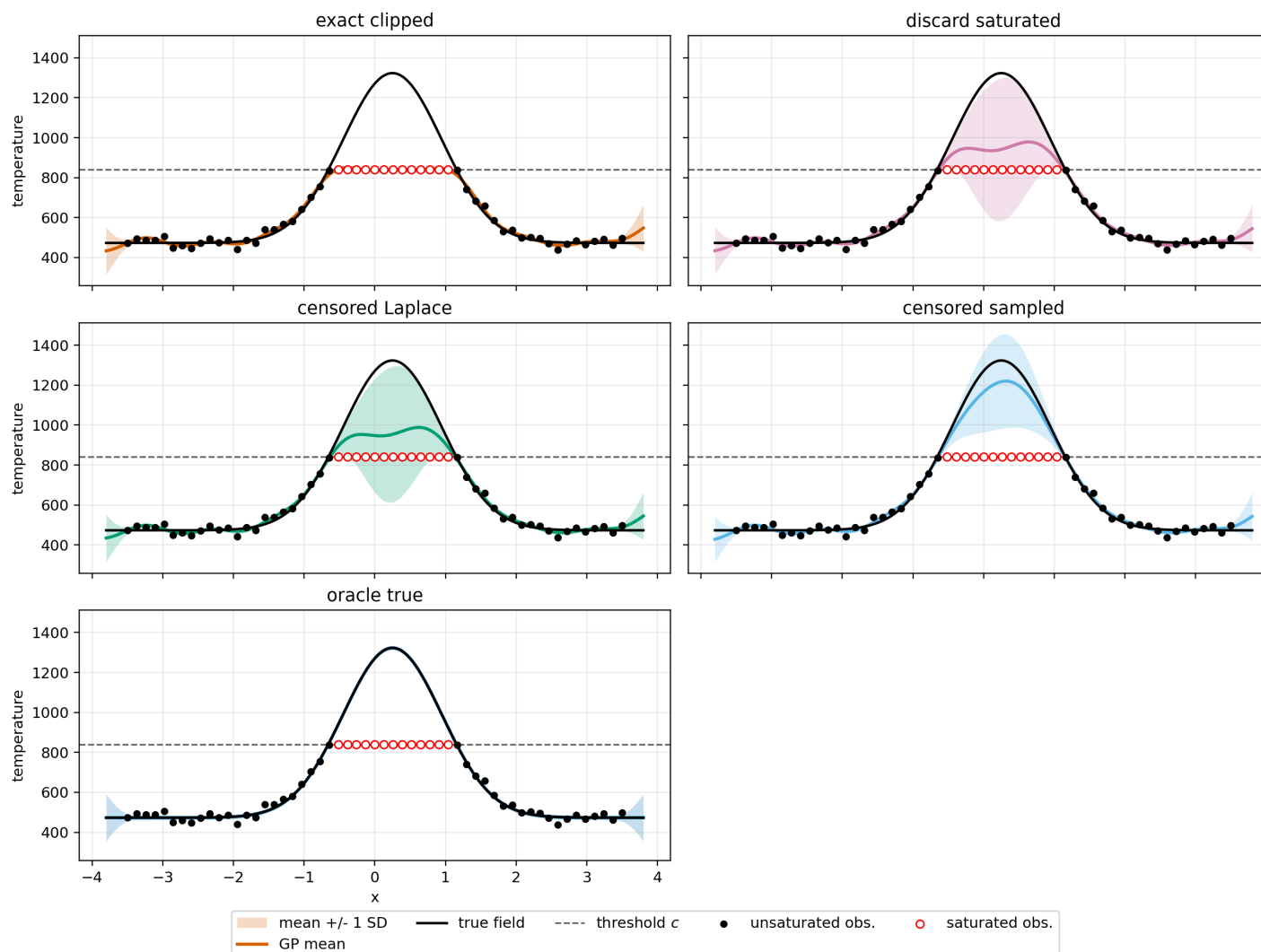
- The censored likelihood helps most when paired with a useful prior. Saturated pixels provide inequality information, but the GP prior decides how that information propagates into the hidden hot region.
- In the 1D fixed-lengthscale test, sampled censored GP reduced peak error from 474 K for clipped regression to 103 K and covered the true peak.
- Unsaturated-only lengthscale tuning was not uniformly beneficial: in 1D, sampled censored coverage fell from 100% with fixed ell to 10% with tuned ell.
- In 2D, isotropic RBF worked reasonably for compact fields but struggled on the moving-laser path; sampled censored peak error was about 637 K with fixed isotropic ell.
- The path-aligned fixed kernel was the clearest improvement for the moving-laser case: sampled censored peak error dropped to 59 K with 100% peak coverage.

Experiment map

Experiment	Kernel/prior used	Main comparison	Main lesson
1D uncertainty bands	1D RBF, fixed ell = 0.55	exact, discard, censored Laplace, censored sampled, oracle	Sampling from the inequality-conditioned posterior gives a much more plausible peak interval.
1D lengthscale sweep/tuning	1D RBF, fixed sweep and unsaturated MLL tuning	ell in {0.3, 0.5, 0.8, 1.2, 1.8}; then 30 seeds	Lengthscale is not a nuisance parameter; it controls whether the hidden peak can rise.
2D shapes and density	Isotropic 2D RBF, ell selected from unsaturated MLL	axis Gaussian, rotated wake, moving-laser path; 13x13, 17x17, 21x21 observations	More observations help, but geometry mismatch remains visible.
2D fixed vs tuned ell	Isotropic 2D RBF, fixed ell = 0.70 vs unsaturated MLL tuned	20 seeds per shape	Unsaturated tuning often selects smoother priors and can worsen hidden-peak recovery.
Kernel misspecification	Isotropic, anisotropic tuned, anisotropic informed, path-aligned	10 seeds, same GP methods	Geometry-aware kernels are essential for wake/path-like fields.

1D Uncertainty Bands

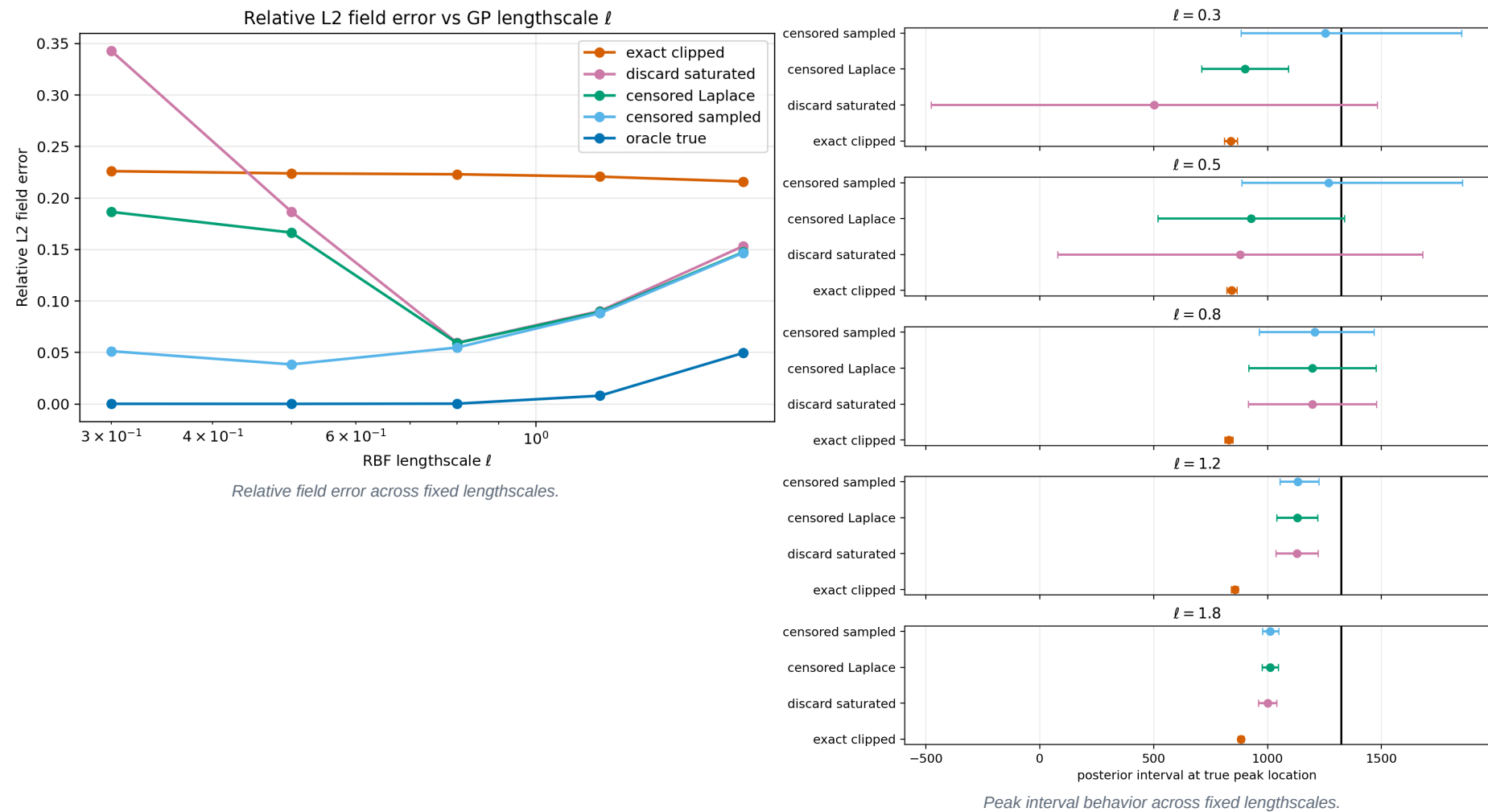
Setup: a 1D Gaussian-like temperature profile was censored by $y_i = \min(T_i + \epsilon_i, c)$. The GP used a squared-exponential/RBF kernel $k(x, x') = \sigma_f^2 \exp(-(x-x')^2/(2\ell^2))$ with fixed $\ell = 0.55$, $\sigma_f = 500$, and noise $sd = 20$. The saturated likelihood contribution is $P(y_i = c | f_i) = \Phi((f_i - c)/\sigma)$, and the sampled censored GP approximates the posterior conditioned on $f_S \geq c$.



One-standard-deviation posterior bands. Exact clipped is sharply biased downward; sampled censored GP gives a broad but plausible hidden-peak range and lowers peak error to 103 K.

1D Lengthscale Sweep

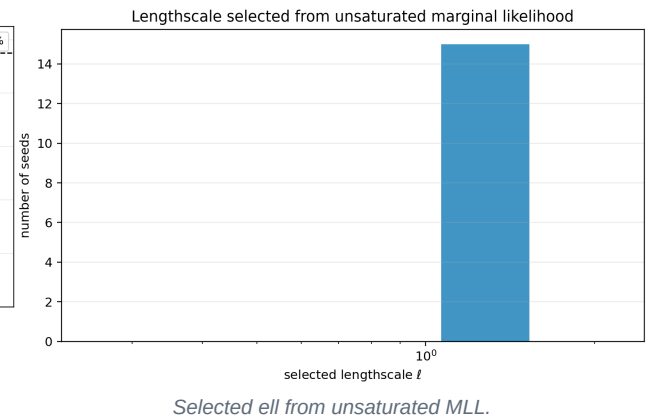
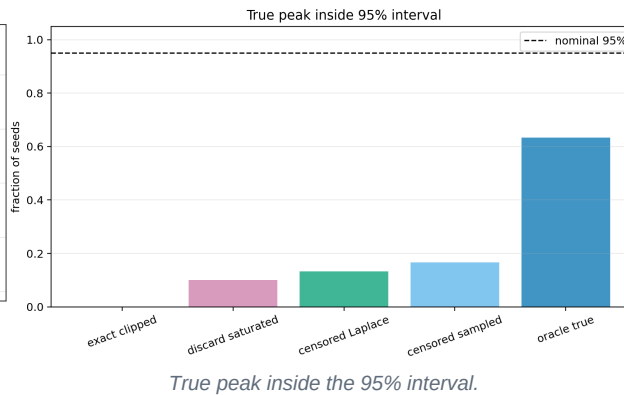
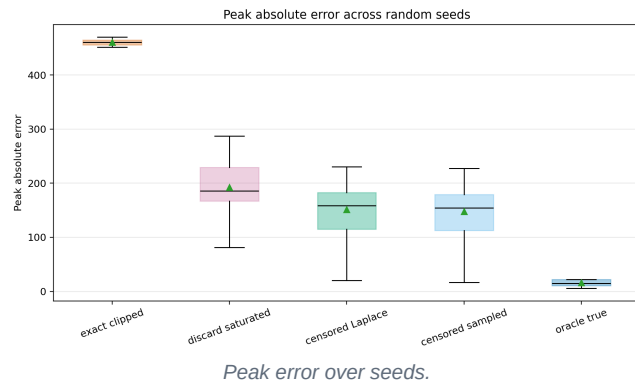
Only the RBF lengthscale ℓ was varied. This tests Adrienne's point that recovering a clipped peak requires a smoothness assumption: if the prior is too local, the interior of the saturated region can stay close to the threshold; if it is too smooth, the model can again underfit the peak.



1D Multi-seed Lengthscale Tuning

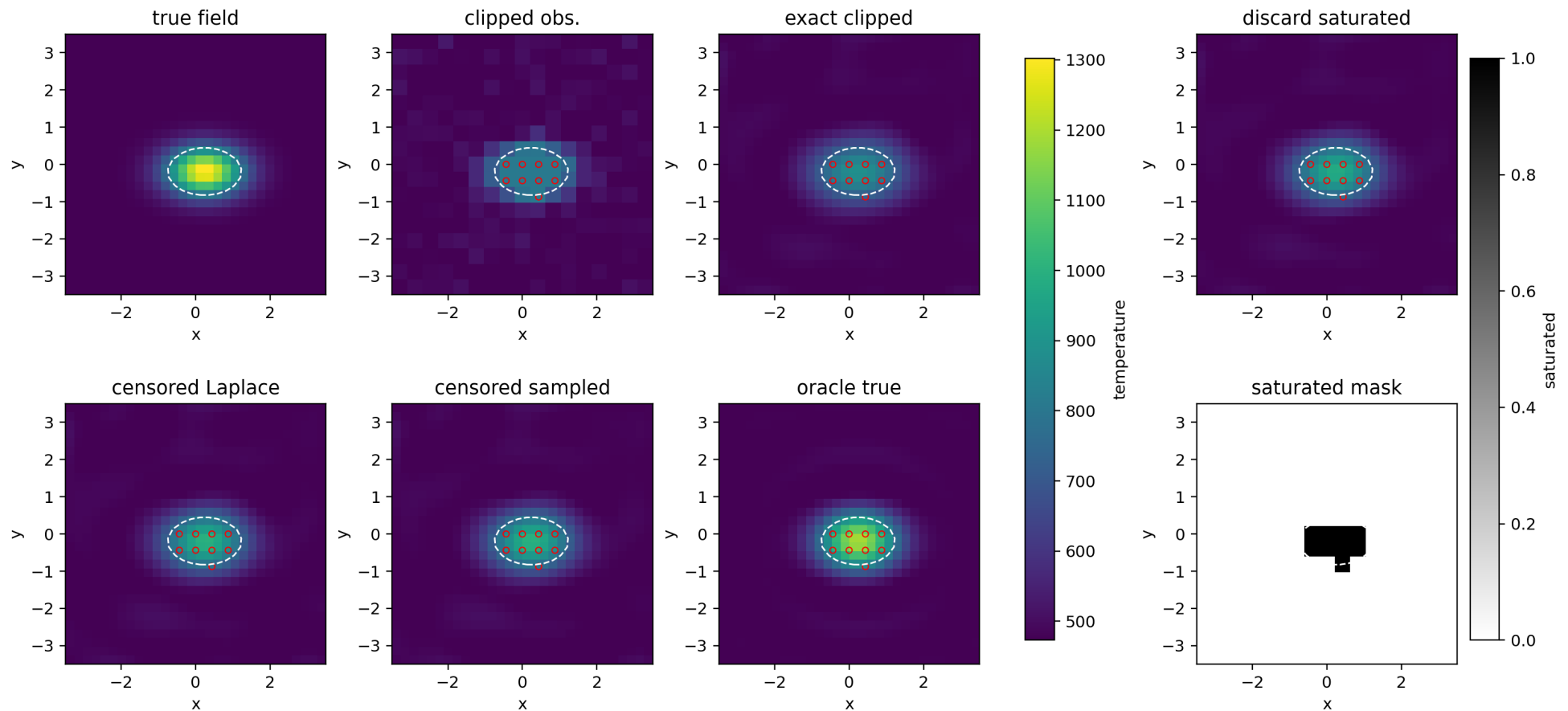
I repeated the 1D experiment over 30 random observation/noise seeds and compared fixed $\text{ell} = 0.55$ with ell selected by maximizing the marginal likelihood of only the unsaturated observations.

Scenario	Method	Field L2	Hot-region L2	Peak error	Peak coverage	Mean ell
fixed ell = 0.55	exact clipped	0.224	0.290	473	0%	0.55
fixed ell = 0.55	discard saturated	0.220	0.284	383	100%	0.55
fixed ell = 0.55	censored Laplace	0.127	0.162	248	83%	0.55
fixed ell = 0.55	censored sampled	0.051	0.058	69	100%	0.55
unsat MLL tuned	exact clipped	0.220	0.284	460	0%	1.27
unsat MLL tuned	discard saturated	0.093	0.119	195	10%	1.27
unsat MLL tuned	censored Laplace	0.073	0.092	154	10%	1.27
unsat MLL tuned	censored sampled	0.072	0.090	151	10%	1.27



2D Shape Study: Axis Gaussian

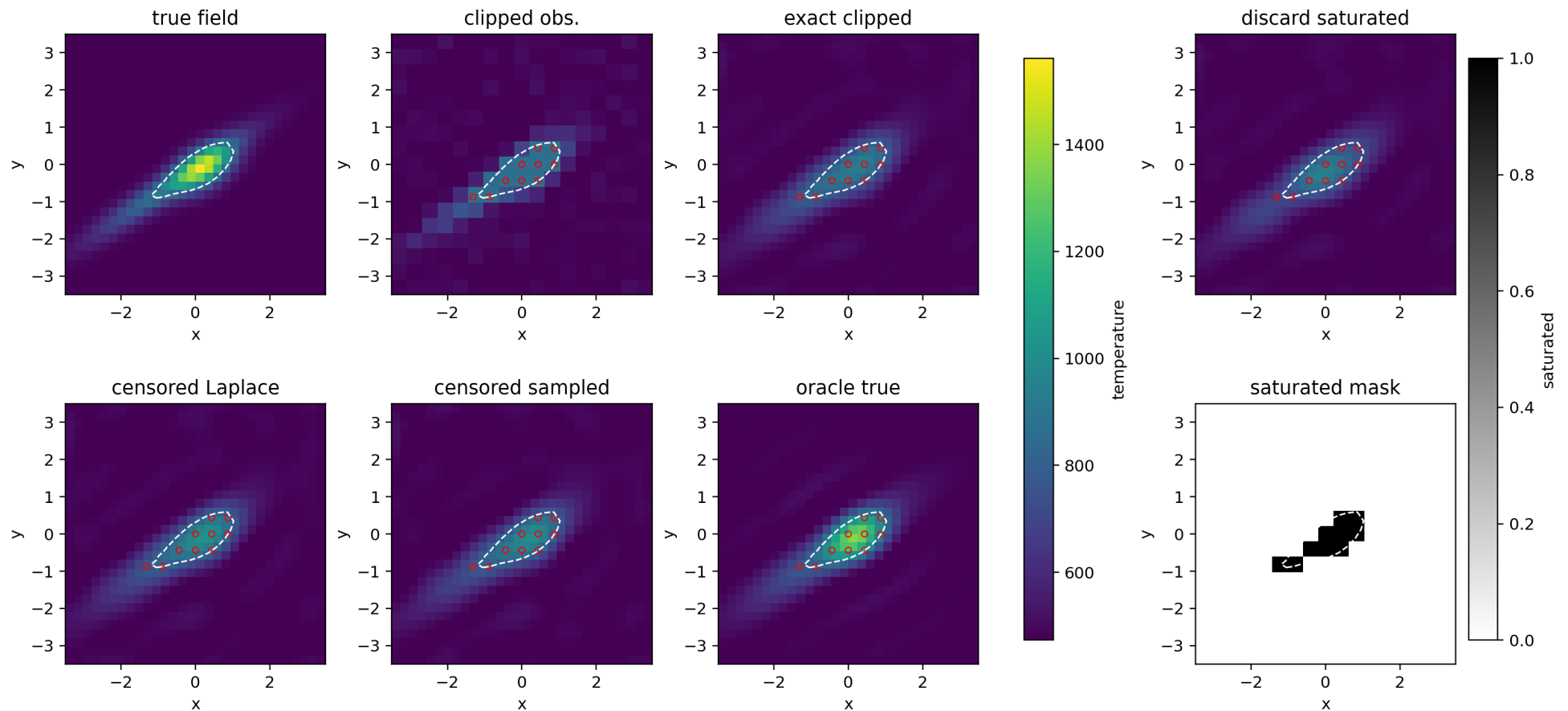
Setup: the first 2D extension used an isotropic 2D RBF kernel $k(z, z') = \sigma_f^2 \exp(-||z - z'||^2 / (2 \ell^2))$. For the shape/density sweep, ℓ was selected from unsaturated observations. The axis Gaussian is the closest 2D analogue of the original toy example.



Axis Gaussian reconstruction examples across observation density. The isotropic RBF prior is well matched to this geometry.

2D Shape Study: Rotated Wake

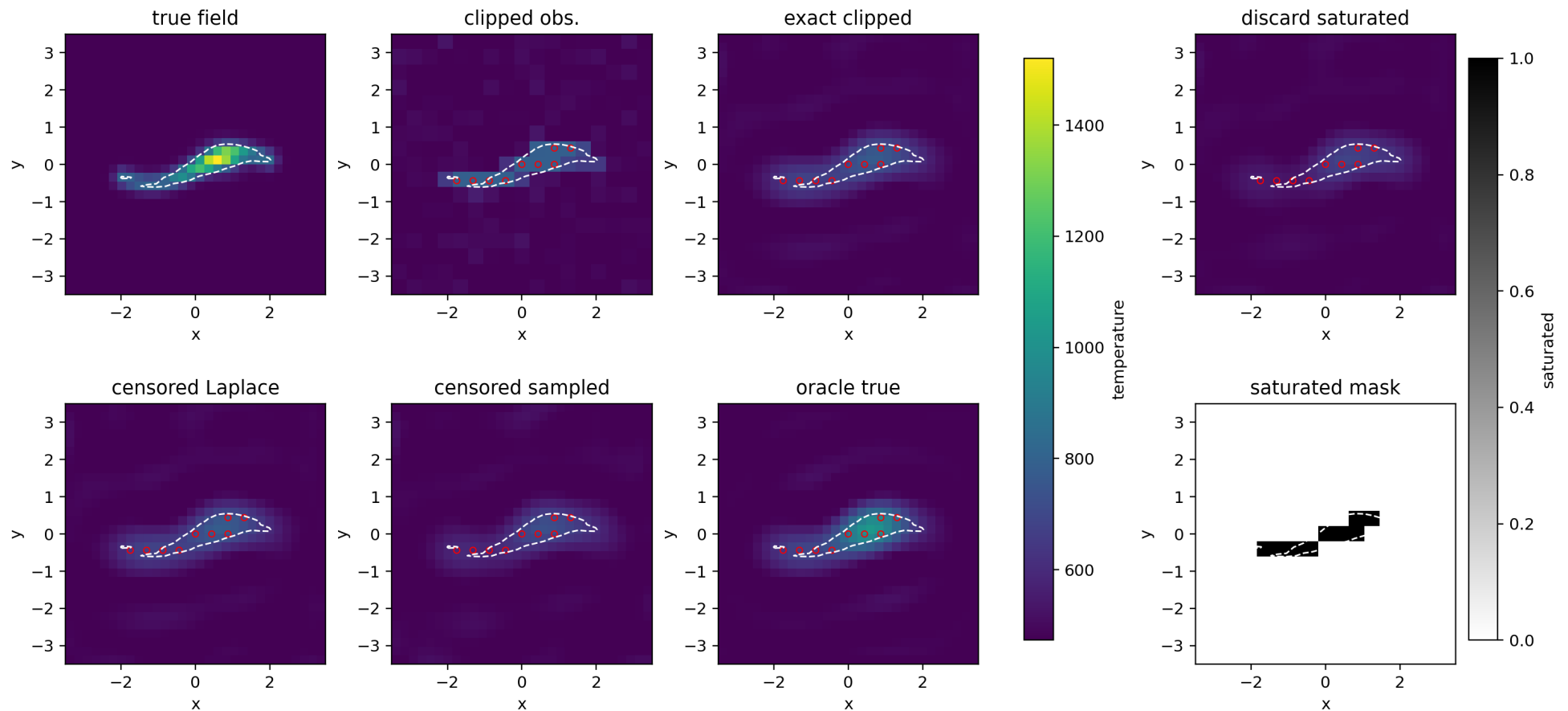
The rotated wake introduces anisotropy and orientation. This is a first test of model misspecification: the isotropic RBF does not encode the long, tilted correlation structure of the field.



Rotated wake reconstruction examples. Censored methods still use the inequality information, but the isotropic kernel has to explain a directionally elongated field.

2D Shape Study: Moving-laser Path

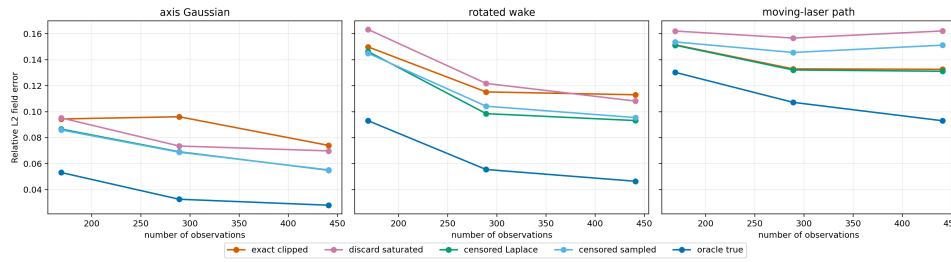
The moving-laser path is the hardest case for a plain isotropic RBF because the high-temperature structure is curved and localized along a path, not radially smooth around one center.



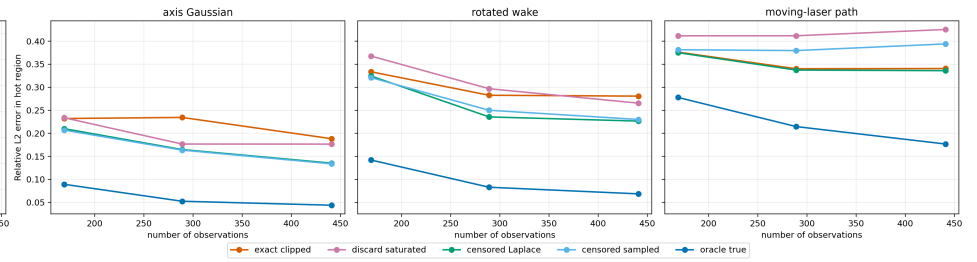
Moving-laser reconstruction examples. The isotropic kernel tends to smear or under-estimate the hidden path peak.

Observation Density

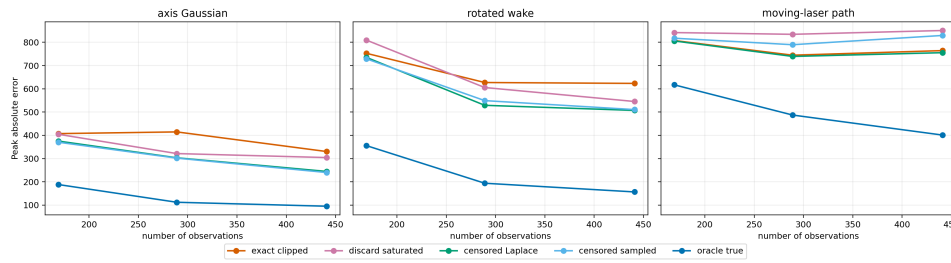
I varied the observation grid size from 13x13 to 21x21 while keeping the same three shapes and the same isotropic RBF setup with unsaturated-data lengthscale selection. This tests whether denser camera sampling alone fixes saturation-induced peak loss.



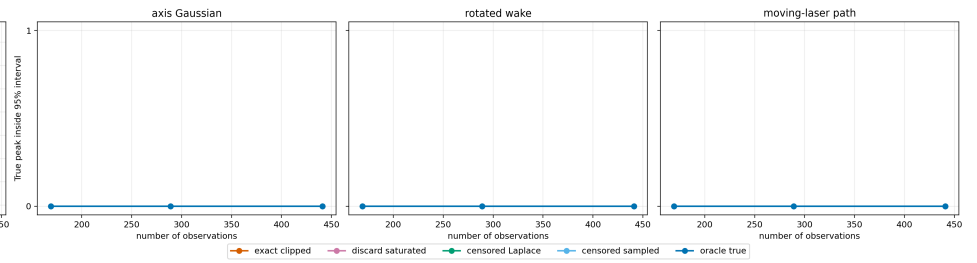
Relative field error versus observation density.



Hot-region error versus observation density.



Peak error versus observation density.

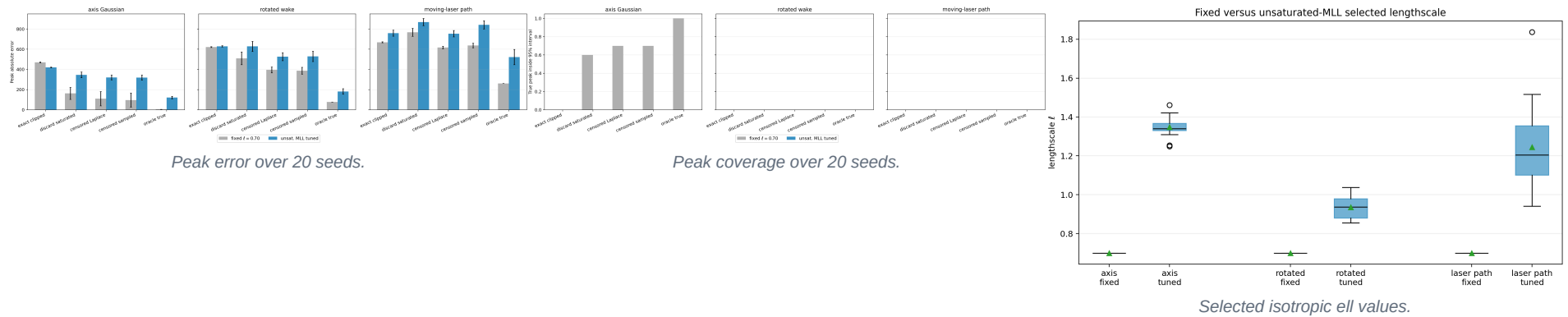


True peak coverage versus observation density.

2D Fixed vs Tuned Isotropic Lengthscale

To separate density effects from hyperparameter effects, I repeated the 2D shape experiment over 20 seeds and compared fixed $\ell = 0.70$ with ℓ selected from unsaturated marginal likelihood.

Shape	Scenario	Sampled field L2	Sampled peak error	Sampled coverage	Mean ℓ
axis Gaussian	fixed $\ell = 0.70$	0.038	97	70%	0.70
axis Gaussian	unsat MLL tuned	0.073	319	0%	1.35
rotated wake	fixed $\ell = 0.70$	0.072	387	0%	0.70
rotated wake	unsat MLL tuned	0.099	529	0%	0.94
moving-laser path	fixed $\ell = 0.70$	0.112	637	0%	0.70
moving-laser path	unsat MLL tuned	0.156	841	0%	1.24

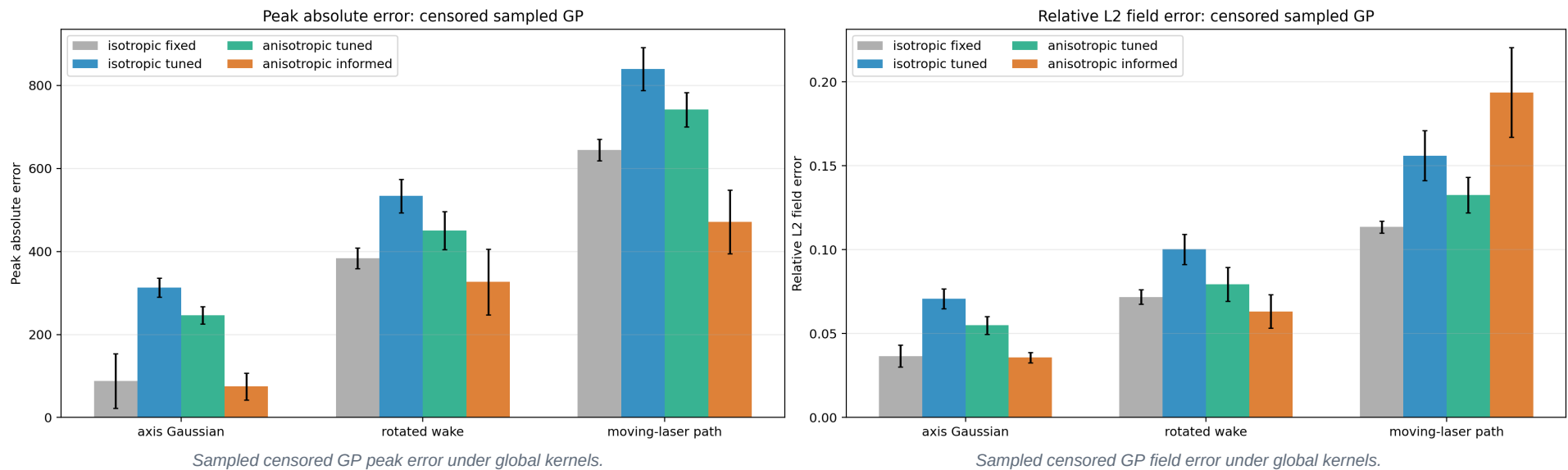


Kernel Misspecification: Isotropic and Global Anisotropic Kernels

I then changed the kernel rather than the observation model. Kernels tested: isotropic fixed $\text{ell} = 0.70$; isotropic tuned by unsaturated MLL; anisotropic tuned over ell_x , ell_y , and angle; and informed anisotropic kernels chosen from the known shape geometry.

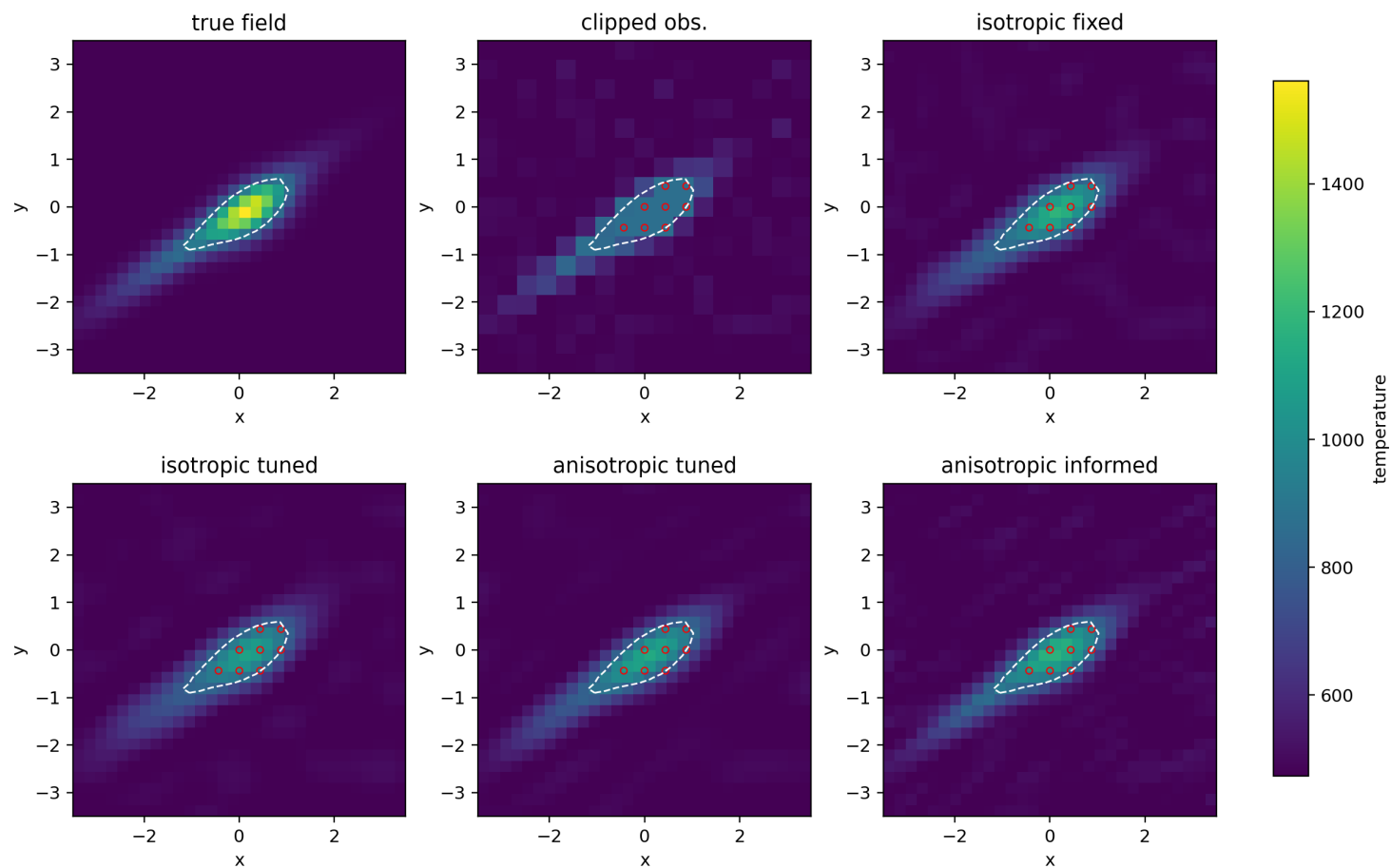
The anisotropic kernel used rotated coordinates (u, v) : $k(z, z') = \sigma_f^2 \exp(-(\frac{u-u'}{\text{ell}_x})^2 - (\frac{v-v'}{\text{ell}_y})^2)$. For the rotated wake, the informed kernel used angle 34 deg, $\text{ell}_x = 1.25$, $\text{ell}_y = 0.28$.

Case	Isotropic fixed sampled peak	Informed anisotropic sampled peak	Coverage change
rotated wake	384	327	0% to 30%
moving-laser path	645	472	0% to 0%



Kernel Misspecification: Reconstruction Example

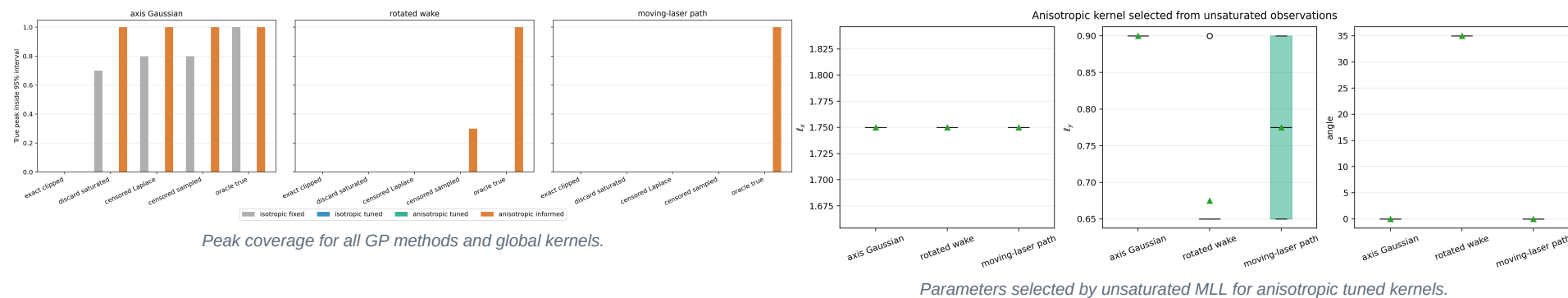
For the rotated wake, the reconstruction panel makes the kernel effect more visible than the error bars alone. The informed anisotropic kernel uses the wake direction and different along-wake/across-wake lengthscales, so the recovered hot region follows the elongated tilted structure more naturally than the isotropic alternatives.



Rotated-wake censored-sampled reconstructions for seed 0 under isotropic and anisotropic global kernels.

Kernel Misspecification: Coverage and Selected Parameters

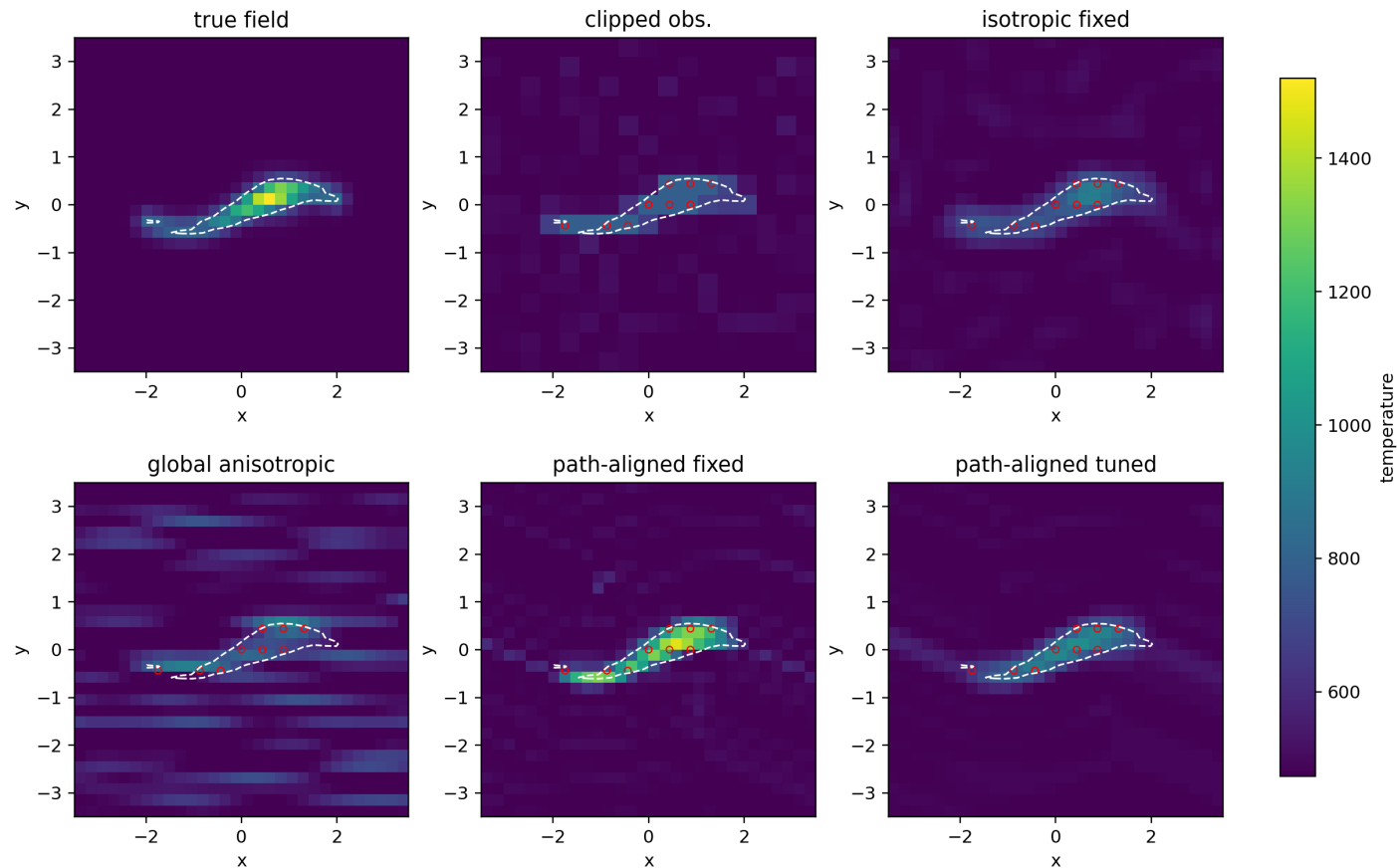
The global anisotropic kernel improved straight or nearly straight anisotropic structure, especially the rotated wake, but it was not enough for the curved moving-laser path. This motivated a kernel whose coordinates follow the laser trajectory.



Path-aligned Kernel for the Moving-laser Field

For the moving-laser field, I added a path-aligned kernel. Each point (x, y) is mapped approximately to path coordinates (s, r) , where s is position along the known laser path and r is signed distance away from it. The kernel is anisotropic in these coordinates: $k = \sigma_f^2 \exp(-(s-s')^2/(2 \ell_s^2) - (r-r')^2/(2 \ell_r^2))$.

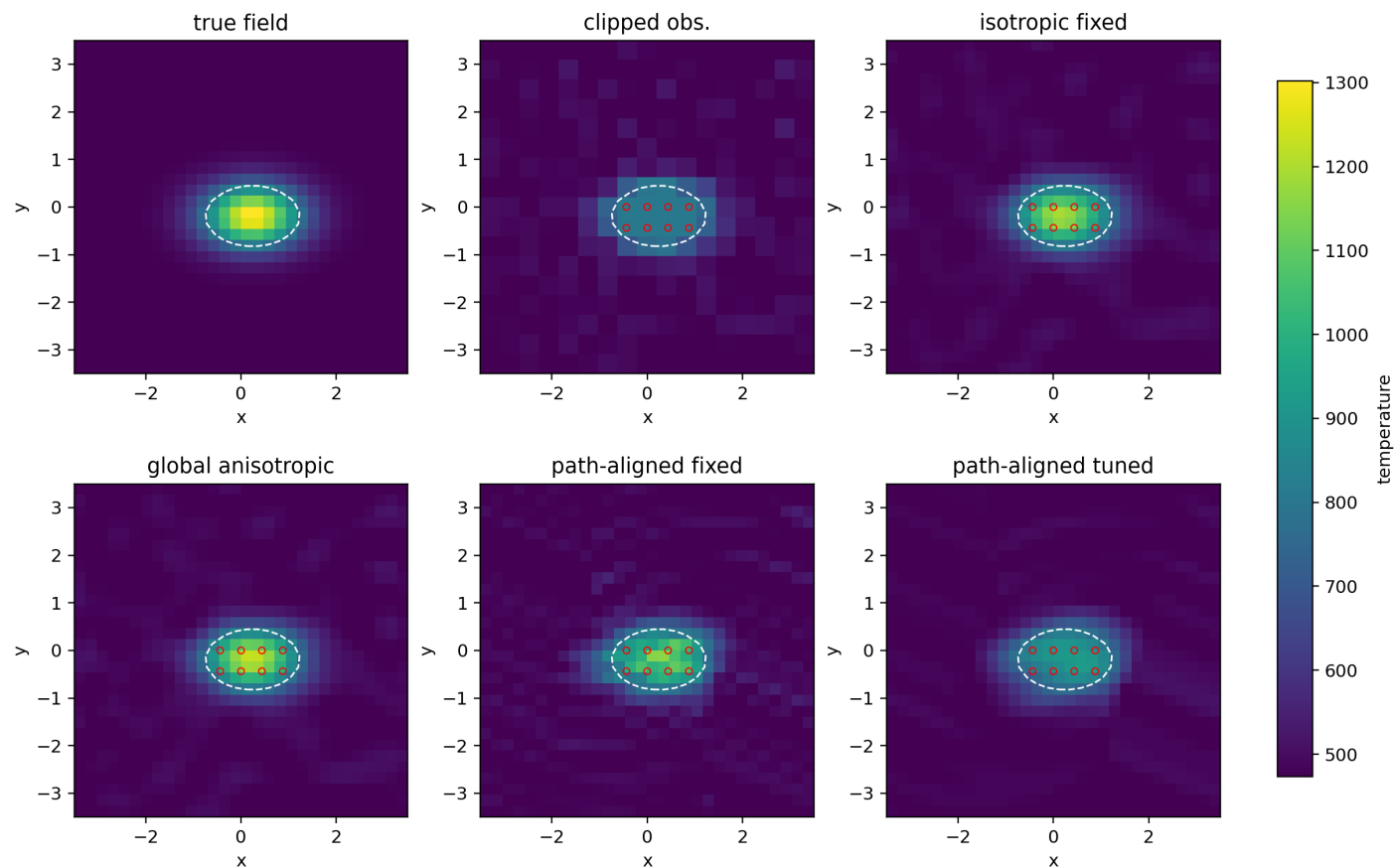
Two versions were tested: path-aligned fixed with $\ell_s = 1.10$ and $\ell_r = 0.20$, and path-aligned tuned with ℓ_s and ℓ_r selected from unsaturated MLL. The tuned version usually chose ℓ_s near 2.05 and $\ell_r = 0.45$, which over-smoothed the hidden peak.



Moving-laser sampled reconstructions for seed 0. The path-aligned fixed kernel is the first prior that reconstructs the hidden peak shape convincingly.

Path-aligned Kernel: Axis Gaussian Check

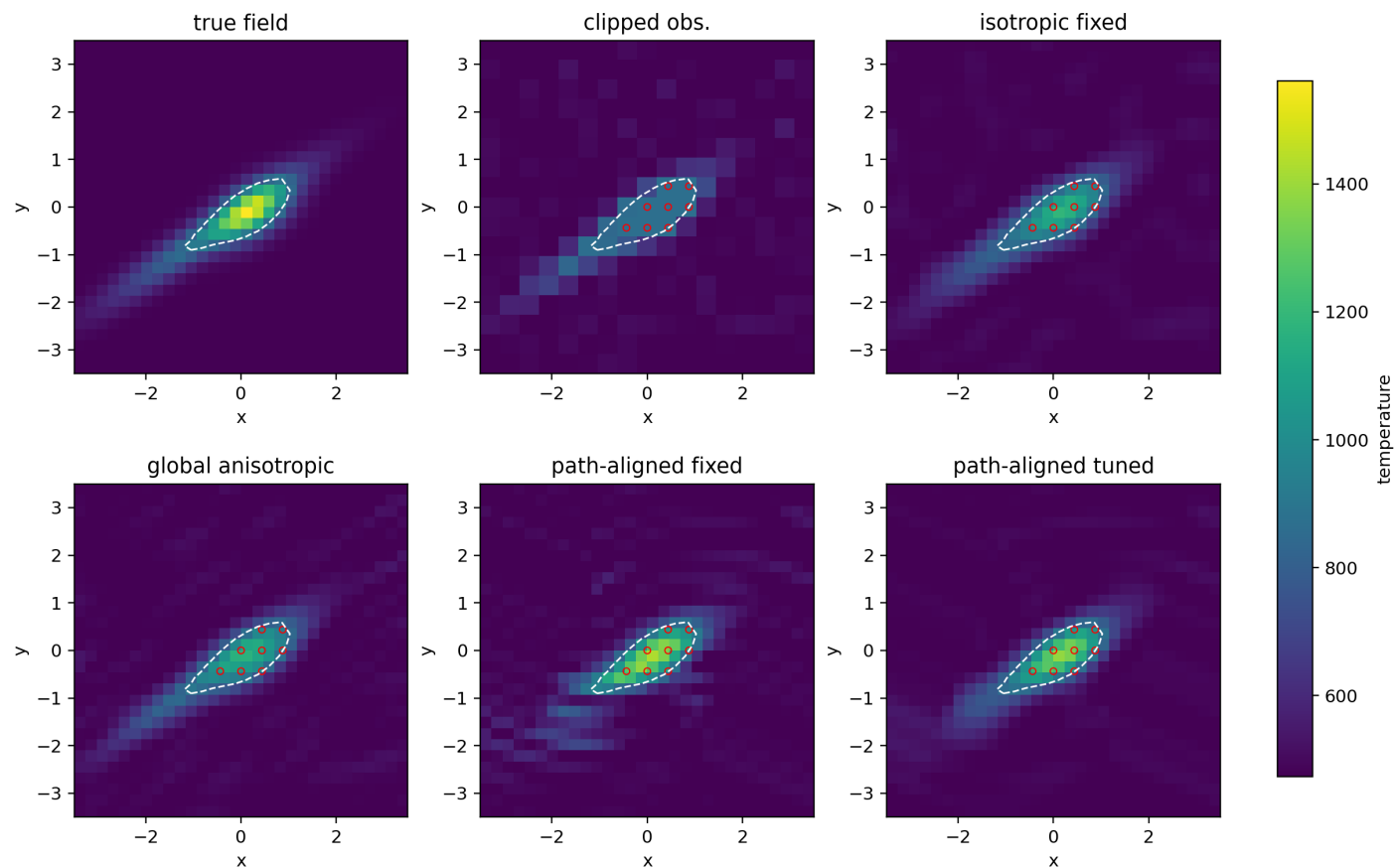
As a sanity check, I also applied the path-kernel comparison panel to the axis Gaussian. This field is not generated by the laser path, so the global anisotropic kernel is the more natural geometry-aware prior here, while the path-aligned panels show what the laser-path prior does when its geometry is not needed.



Axis-Gaussian censored-sampled reconstructions for seed 0 under isotropic fixed, global anisotropic, path-aligned fixed, and path-aligned tuned kernels.

Path-aligned Kernel: Rotated Wake Check

The rotated wake is useful as a misspecification check: it has a preferred direction, but not the curved laser-path geometry. This panel helps separate the value of generic anisotropy from the value of a path-specific kernel.

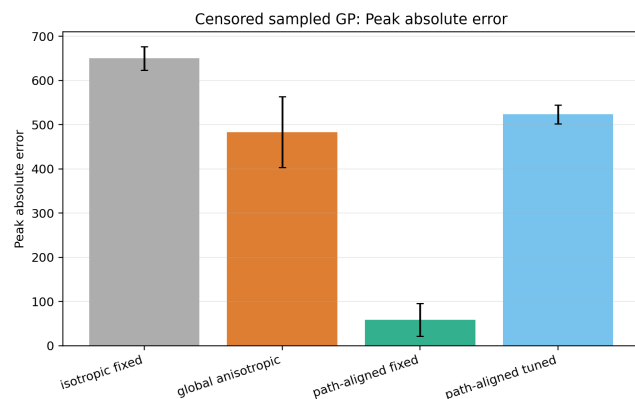


Rotated-wake censored-sampled reconstructions for seed 0 under isotropic fixed, global anisotropic, path-aligned fixed, and path-aligned tuned kernels.

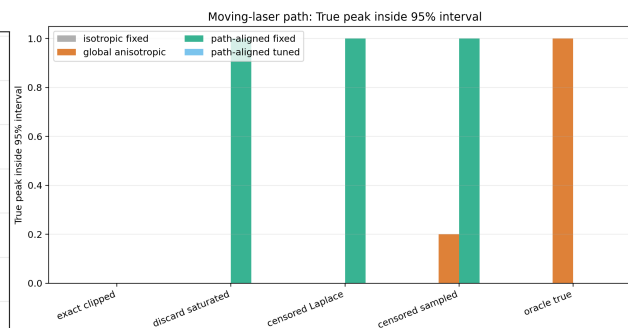
Path-aligned Kernel Results

For censored sampled GP, the path-aligned fixed kernel reduced mean peak error from 650 K under isotropic fixed to 59 K, and peak coverage improved from 0% to 100%. The tuned path-aligned kernel had lower global field error but missed the peak because the unsaturated data favored an overly smooth path prior.

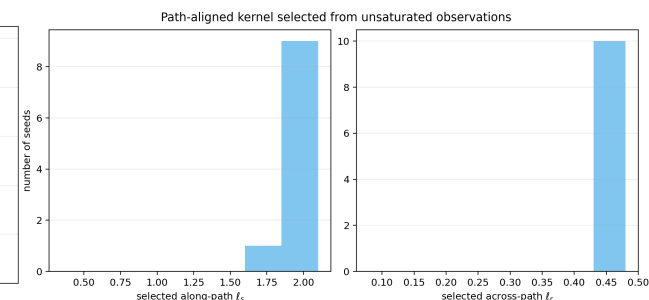
Kernel	Field L2	Hot-region L2	Peak error	Peak coverage	ell_s	ell_r
isotropic fixed	0.114	0.290	650	0%	0.70	
global anisotropic	0.201	0.274	483	20%	1.10	0.20
path-aligned fixed	0.103	0.209	59	100%	1.10	0.20
path-aligned tuned	0.088	0.216	523	0%	2.05	0.45



Sampled censored GP peak error.



Peak coverage for all methods.



Path-kernel parameters selected from unsaturated MLL.

Interpretation and Next Numerical Step

- The important variable is now the prior/kernel, not just the censoring likelihood. Censoring tells the model that hidden values exceeded the camera ceiling; the kernel decides how high and where that hidden mass can plausibly be.
- Exact clipped GP remains a useful negative control: it is consistently biased downward and overconfident near the saturated peak.
- Discarding saturated pixels can be competitive when the prior is already strong, but it wastes direct inequality information and can create overly broad or shape-dependent uncertainty.
- Laplace censored GP is useful as a cheap approximation, but sampled censored GP is more informative for peak uncertainty because the inequality-conditioned posterior can be skewed.
- Unsaturated-only MLL is not enough for hyperparameter selection in this problem. It often sees the smooth cool region well but cannot see the hidden hot peak.
- The most promising direction is geometry-aware censored GP: use kernels informed by the heat source path, wake direction, material anisotropy, or a simple physics model, then select hyperparameters using the censored likelihood or validation on synthetic fields.

Candidate next experiment

Use the path-aligned kernel as the main moving-laser baseline, then compare fixed geometry-informed hyperparameters against hyperparameters learned with the full censored likelihood rather than only unsaturated observations. This directly tests whether the saturation-aware likelihood can improve both reconstruction and calibration when the kernel family is appropriate.